



Overview This project demonstrates the setup and configuration of a virtualized network environment using two virtual machines — one Linux-based and one Windows-based. The goal is to practice fundamental system administration, networking, and basic security hygiene concepts in a controlled lab environment.

1) Lab Environment

- **Hypervisor:** VirtualBox
- **Virtual machines:**
 -  **WIN-CLIENT-01** (Windows)
 -  **LINUX-SERVER-01** (Linux)

2) Network Design / Configuration

- **NAT** for internet access

3) Connectivity Validation

- Confirmed communication between **WIN-CLIENT-01** and **LINUX-SERVER-01** over the internal network.
- **Internal Network** for VM-to-VM communication (sent pings)

4) Basic System Hardening

Windows (WIN-CLIENT-01)  

- Enabled **Windows Defender**
- Configured **Windows Firewall** rules

Linux (LINUX-SERVER-01) 🔥🔒

- Enabled and configured **UFW** firewall
- Hardened **SSH** configuration
- Installed and configured **Fail2Ban**

5) Snapshots and Rollback 🗑️🔙

- Created **snapshots** to save clean states and support quick rollback after testing or configuration changes.

6) Key Takeaways 🧠🔒

- Build a **safe testing environment**
- Apply **security hardening** early
- Keep everything **documented and repeatable**